Angular HandsOn 5: Reactive Forms and Validation

HANDS-ON 5 [Intermediate]

Reactive Forms ? FormBuilder, FormGroup, FormArray & Custom Validators

Topics Covered:

ReactiveFormsModule Setup[Check] FormBuilder & FormGroup[Check]

FormControl & FormArray[Check] Built-in Validators[Check]

Custom Synchronous Validators[Check] Custom Async Validators[Check]

Dynamic Form Controls[Check]

Reactive forms define form structure in the component class rather than the template ? making them more powerful, testable, and suited t- complex scenarios. You will rebuild the enrollment form as a reactive form and add custom validators and dynamic controls.

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Add ReactiveFormsModule t- the component's imports array (standalone) or the module's imports t- enable FormBuilder, FormGroup, and FormControl.

Task 1: Build a Reactive Form with FormBuilder

Goal: Define and bind a reactive form using FormBuilder.

Steps:

48. Generate a ReactiveEnrollmentFormComponent: ng generate component pages/reactive-enrollment-form. Add a route at /enroll-reactive.

49. Inject FormBuilder in the constructor: constructor(private fb: FormBuilder) {}. In ngOnInit, build the form: this.enrollForm = this.fb.group({ studentName: ['', [Validators.required, Validators.minLength(3)]], studentEmail: ['', [Validators.required, Validators.email]], courseId: [null, Validators.required], preferredSemester: ['Odd', Validators.required], agreeToTerms: [false, Validators.requiredTrue] });

50. Bind the form in the template using [formGroup]='enrollForm'. Bind each input using formControlName='studentName' (etc.). N- ngModel is needed in reactive forms.

51. Add a Submit button: <button [disabled]='enrollForm.invalid'>Submit</button>. On submit, log enrollForm.value and enrollForm.getRawValue().

52. Explain in a comment the difference between enrollForm.value (excludes disabled controls) and enrollForm.getRawValue() (includes all controls).

Hint:

- In reactive forms, the form model lives entirely in TypeScript ? the template just binds t- it. This makes the form logic fully unit-testable without a DOM.

- Validators.requiredTrue specifically validates that a checkbox is checked ? Validators.required only checks for a non-empty/non-null value.

Expected Outcome: Form renders correctly bound t- the reactive model. Submit logs the form value. Disabled

Submit button enables only when all validators pass.

Task 2: Custom Validators and FormArray for Dynamic Controls

Goal: Write a custom synchronous validator and use FormArray for repeating controls.

Steps:

53. Write a custom validator function noCourseCode(control: AbstractControl): ValidationErrors | null that returns { noCourseCode: true } if the control value starts with 'XX' (a disallowed prefix). Apply it t- the courseId control alongside Validators.required.

54. Display the custom error: <span *ngIf='enrollForm.get("courseId")?.errors?.["noCourseCode"]'>Course code starting with XX is not allowed..

55. Write an async validator simulateEmailCheck that returns a Promise: after 800ms, if the email contains 'test@', return { emailTaken: true }, otherwise return null. Apply it as the third argument of the email FormControl: this.fb.control('', [Validators.required, Validators.email], [simulateEmailCheck]).

56. Add a FormArray for additionalCourses: this.fb.array([]). Add an 'Add Another Course' button that pushes a new FormControl('', Validators.required) int- the array. Render the array: <div *ngFor='let ctrl of additionalCourses.controls; let i=index'><input [formControl]='ctrl'><button (click)='removeCourse(i)'>Remove</button> .

57. Implement get additionalCourses() { return this.enrollForm.get('additionalCourses') as FormArray; } as a typed getter ? explain why this getter is better than casting in the template.

Hint:

- Async validators fire after all sync validators pass (t- avoid unnecessary API calls). They return Observable<ValidationErrors | null> or Promise<ValidationErrors | null>.
- FormArray is Angular's answer t- dynamic, repeating form sections ? like adding multiple phone numbers or, in this case, enrolling in multiple courses at once.

Expected Outcome: Entering 'XX101' in courseId shows the custom error. test@example.com shows 'Email taken' after 800ms. Additional course controls can be added and removed dynamically.

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